GPS-denied high-tunnel UAV

Preliminary configuration and quote basis

Prepared for UAV-IQ · UCLA high-tunnel biocontrol project

5 November 2026

Preliminary estimates, subject to the open inputs on slide 13

Key points

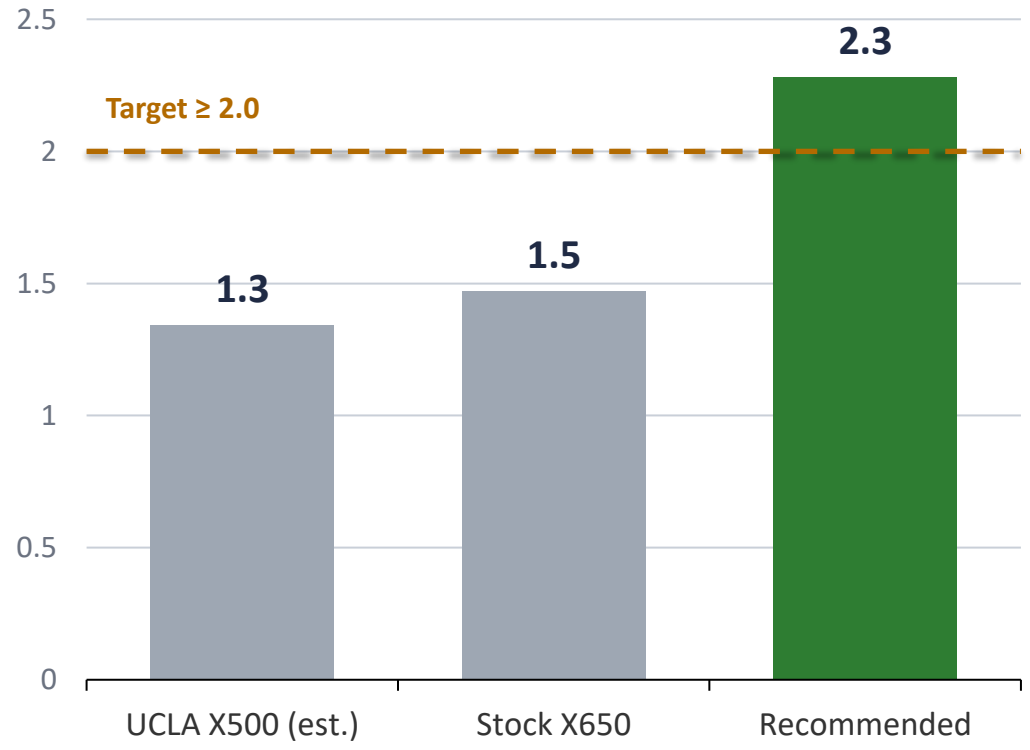
- **Within our capabilities.** A propulsion, payload-power and sensor-integration build on UCLA's existing PX4 + Jetson stack.
- **Current options fall short.** The X500 lacks thrust margin with the releaser; a stock X650 copes with 1 L but has no margin for 2 L or 4 L; the Starling 2 Max cannot carry it.
- **Recommendation.** An X650-class quad on 6S with uprated motors, reusing UCLA's Pixhawk, Jetson and D455, and adding flow and range sensors, a scanning lidar and a 12 V blower supply.
- **Performance.** With the 1 L releaser: thrust-to-weight ≈ 2.3 and ≈ 15 min of releasing. In simulation it holds height through hard dodges that drop the X500 by 0.56 m. 2 L is feasible with less margin; 4 L is not.
- **Cost.** Base package ≈ $11,400–16,500: hardware plus US-based assembly and integration. Flight-stack software set-up is optional. Six inputs from you set the firm quote.

The X500 runs out of thrust with the releaser

- UCLA's X500 with its autonomy kit, a 4S 5 Ah pack and the 1 L dummy weighs ≈ 3.4 kg.
- Thrust-to-weight ≈ 1.3 at nominal battery voltage (≈ 1.6 on a full pack); it hovers at ≈ 80 % throttle.
- ≈ 5–6 min of usable flight with a 20 % reserve.
- Little margin is left for corrections, which matches the altitude dips, wobble and close passes seen in the trials.
- A stock X650 does better (≈ 1.5) and copes with 1 L in our simulation, but has no margin for 2 L or 4 L. Its 4.3 kg payload rating assumes no battery and 70 % hover throttle.

X500 figures are estimates. UCLA's weighed take-off mass and flight logs will confirm them.

Thrust-to-weight with the 1 L releaser



Maximum thrust ÷ take-off weight at nominal battery voltage (3.7 V per cell).

Starling 2 Max: strong navigation, but it cannot carry the releaser

Strengths

- Mature visual-inertial navigation (VOXL 2 computer, dual fisheye tracking cameras)
- Small: about 0.5 m tip to tip
- 40–55 min of flight without payload

Limits for this mission

- Rated payload 500 g; the 1 L releaser alone is 1,073 g (2.1×), before its mount
- 43 Wh battery: the 60 W blower alone would drain it in ≈ 43 min, before any flight power
- Different computer (Qualcomm QRB5165): UCLA's Jetson models would need porting
- Lead time ≈ 60 business days

Useful as a navigation-only test aircraft; not suited to the combined navigation-and-payload mission.

Recommended: X650-class quad on 6S, reusing UCLA's autonomy core

Reuse from the X500

- Pixhawk 6X + Jetson Orin NX baseboard (with NVMe and Wi-Fi)
- RealSense D455 depth camera
- GNSS/compass (outdoor backup only)
- Telemetry radio and RC transmitter
- UBEC 12A (Jetson power)

Replace

- Frame and motors → X650 frame, T-Motor MN4014 KV400 motors, 45 A ESCs, 16×5.4 carbon props
- 4S packs → 6S 10 Ah LiPo (×3)
- Optical-flow sensor → ARK Flow (flow + downward rangefinder)

Add

- Upward rangefinder (roof and lights)
- LightWare SF45/B scanning lidar → PX4 collision prevention
- 12 V / 10 A blower supply, switched by the flight controller, with an RC cut-off
- Quick-release releaser mount
- Remote ID module (if flown outdoors between tunnels)

Why: more than twice the X500's thrust, room for the releaser, and no change to UCLA's software stack.

Estimated performance: 1 L with margin, 2 L feasible, 4 L not

Payload (incl. mount)	Take-off weight	Thrust-to-weight	Flight time, navigating	Flight time, releasing	Verdict
None (navigation milestone)	4.0 kg	3.0	22–27 min	–	**Yes**
1 L: 1.27 kg	5.3 kg	2.3	16–18 min	14.5–16 min	**Yes, recommended**
2 L: 2.10 kg	6.1 kg	2.0	13–14 min	12–13 min	**Yes, reduced margin**
4 L: 2.99 kg	7.0 kg	1.7	10–11 min	≈ 10 min	**No: needs a larger aircraft**

≈ 2.0 kg

payload capacity after the autonomy hardware (thrust-to-weight 2.0, mount included)

120 W for 60 W

12 V blower supply vs. the blower's peak draw (50 % headroom)

≈ 1.06 m

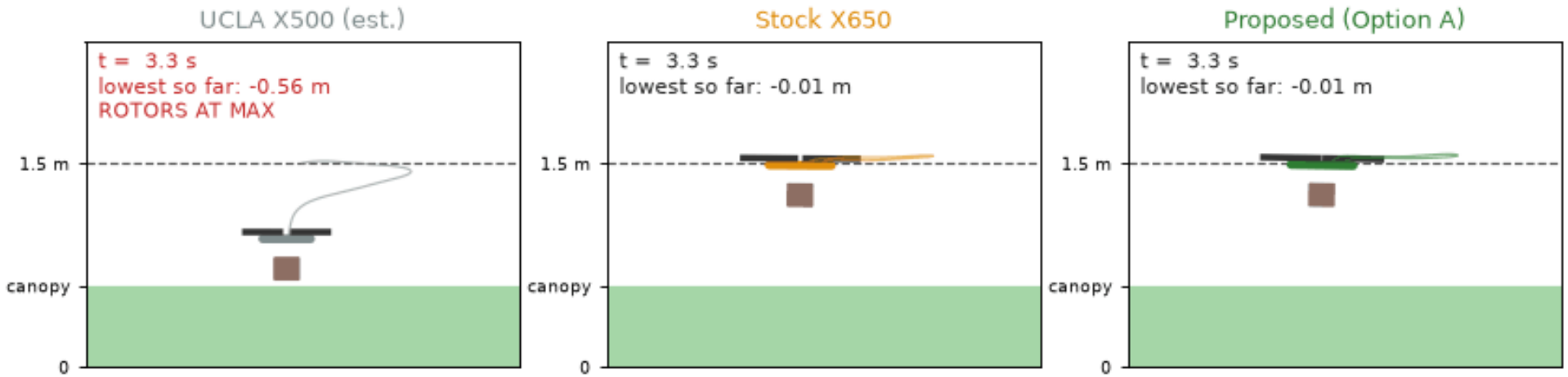
tip-to-tip size: must fit the tunnel envelope

6S 10 Ah LiPo, landing with a 20 % reserve. Based on manufacturer thrust-stand data and confirmed by weighing parts during the build. Mount mass assumed 200–300 g.

How the new aircraft addresses the trial issues

Seen in UCLA's trials	What changes
Altitude dips during corrections	Thrust-to-weight ≈ 2.3 instead of ≈ 1.3; height held with a downward rangefinder
Close passes and manual corrections	Scanning lidar and upward rangefinder feed PX4 collision prevention, which stops the aircraft even if the Jetson stalls
Wobble in hover and slow flight	Lower hover throttle, vibration-isolated avionics, larger airframe
1–2 m landing error	Precision landing on a printed fiducial pad with a downward camera
Dust, sunlight and prop wash	Cameras mounted out of the rotor wash; IR-pass or IP65 camera options; ≈ 22 % lower downwash speed from larger props

Simulation: two hard dodges on a 30 % battery

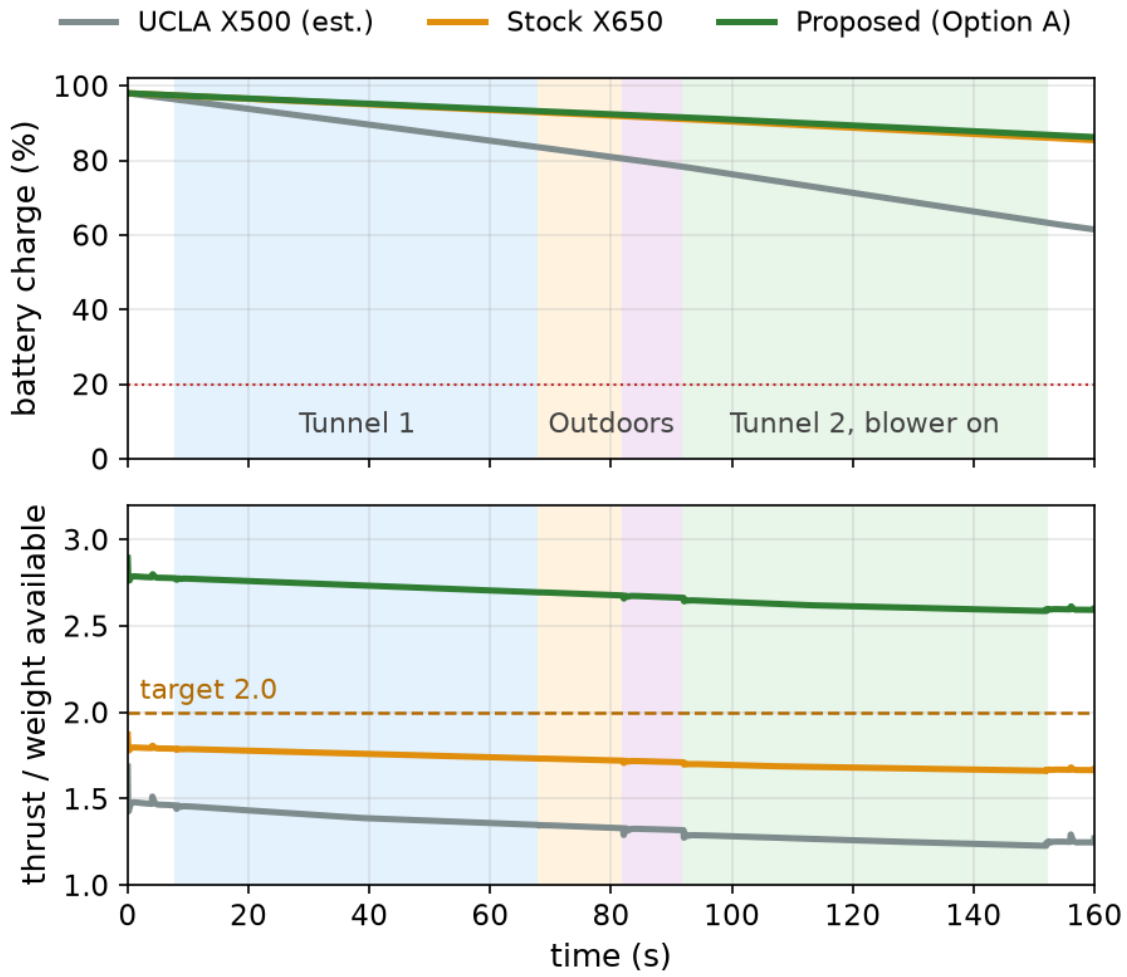


Deepest point of the UCLA X500 (est.) dip (t = 3.3 s). The animation follows.

Same stick input, same controller: the X500 sinks 0.56 m with its rotors at maximum 58 % of the time. The proposed aircraft holds height within 3 cm and never runs out of thrust.

Side view with the 1 L releaser: two ±30° dodges in altitude-hold mode. Our 2-D flight model with manufacturer motor data; inertia, motor lag and battery resistance are estimates. With a full battery all three hold height.

Simulation: a two-tunnel sortie with the 1 L releaser

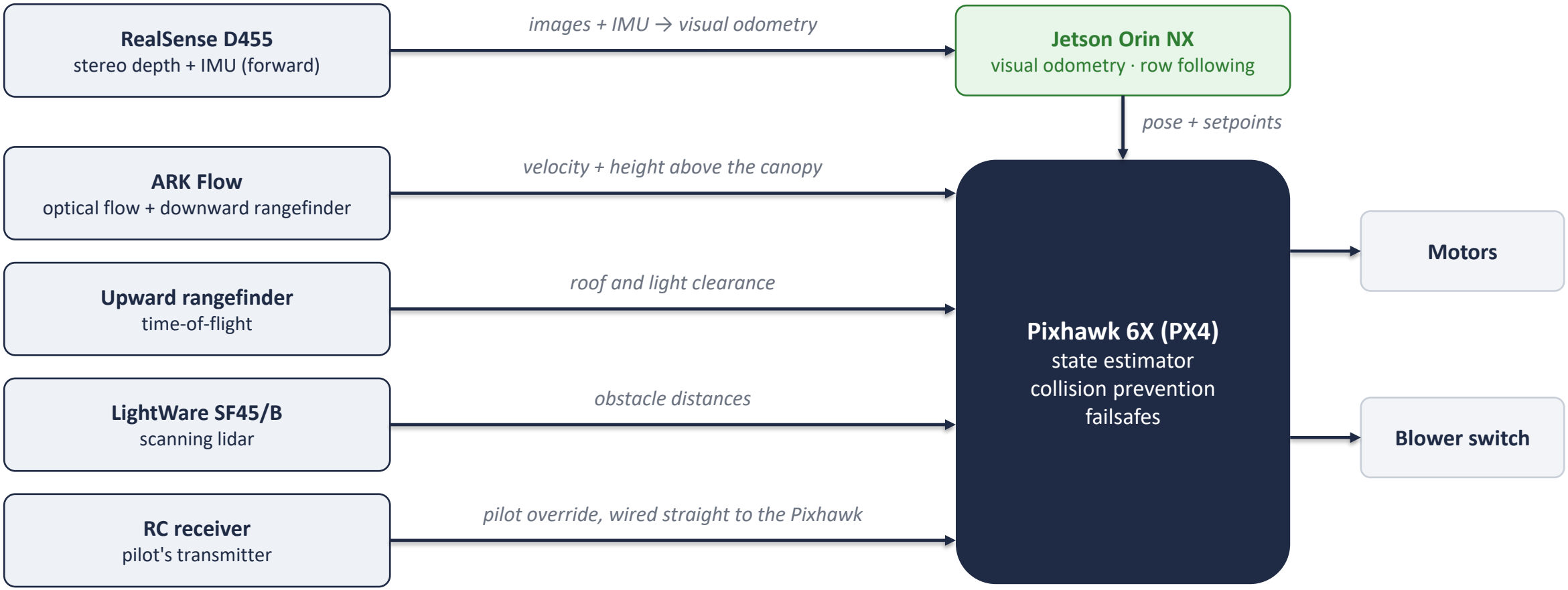


Per sortie	X500 (est.)	Stock X650	Proposed
Battery used	37 %	13 %	**12 %**
Sorties per charge	2.1	5.9	**6.3**
Thrust-to-weight	1.23 → 1.12	1.66 → 1.45	**2.59 → 2.21**
Peak motor load	96 → 100 %	74 → 87 %	**47 → 56 %**

- Three times the X500's sorties per charge
- Thrust-to-weight stays above 2.2 down to the reserve. On this gentle route the stock X650's motors reach 87 % of maximum, ours 56 %

Sortie: 60 m tunnel, 20 m windy outdoor leg, 10 s reorientation, 60 m tunnel with the blower on (160 s). Sorties flown back to back from 98 % charge to a 20 % reserve; arrows run from the first sortie to the last full one. Estimates for comparison.

Navigating without GPS



Heading comes from visual odometry because the steel hoops disturb the compass. GNSS is optional outdoors between tunnels. Recovery uses precision landing on a printed fiducial pad.

Pilot takeover and safety

Pilot authority

- RC receiver wired directly to the Pixhawk: works even if the Jetson stops
- One switch hands control from autonomy back to the pilot (Position, Altitude or Stabilized mode)
- Dedicated kill switch and a physical arming switch

Automatic protection

- Failsafes for RC loss, companion-computer loss, low battery and estimator faults: hold or land
- Geofence on a local origin, so it works without GPS
- Blower power on its own RC cut-off; anti-spark battery connector
- Options: propeller guards (≈ 300 g), Remote ID broadcast module

Preliminary quote

Item	Estimate
Hardware: new parts (≈ $3,700 at list) plus shipping, import duties and procurement	**≈ $4,600**
Engineering: integration, sensor bring-up, safety set-up, documentation (30–53 h at $175/h)	**$5,250–9,275**
Technician: assembly, wiring, mount fabrication (16–27 h at $95/h)	**$1,520–2,565**
Base package (UCLA's autonomy core reused)	**≈ $11,400–16,500**
Option: flight-stack set-up: PX4 sensor fusion, collision prevention, failsafes, Jetson drivers (40–60 h)	**+ $7,000–10,500**
Option: builder hover check and baseline tune, if you allow flights (4–8 h)	**+ $700–1,400**
Option: second autonomy set so the X500 keeps flying (Jetson bundle, D455, GNSS, radio)	**+ ≈ $3,600**

Assembled and integrated in the USA; most components are imported. Excludes autonomy software development, flight and field testing, organisms, efficacy evaluation and sales tax.

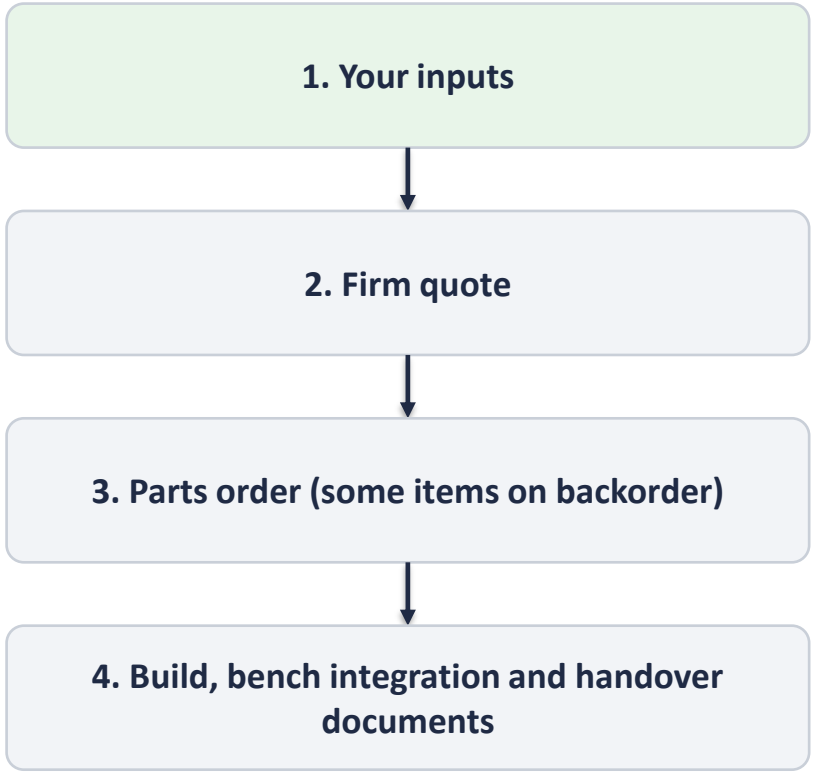
List prices seen 5 November 2026. Lower-cost and 3D-lidar variants on request. The firm quote follows the inputs on the next slide.

What we need to finalize the quote

1. **Tunnel envelope:** width, height, door size, bed spacing, and heights of supports, irrigation and lights (the aircraft is ≈ 1.06 m wide)
2. **Reuse or new:** will the X500's Pixhawk/Jetson, D455, GNSS and radios move to the new aircraft?
3. **Software:** hardware only, or also a configured flight stack (PX4 sensor fusion, failsafes, Jetson drivers)? Autonomy development stays with UCLA unless you want our support.
4. **Funding:** are federal funds involved? Federal rules on foreign-made drone components could change the parts list.
5. **Releaser:** mass, size, mounting points, 12 V inrush and duty cycle, control signal
6. **Acceptance:** bench-verified delivery, or a short builder hover and tune?

Also helpful: the X500's weighed take-off mass and flight logs, the endurance target, peak tunnel temperature, and the plan for the outdoor leg (Part 107, Remote ID).

Next steps



References

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[7] Holybro, X500 V2 motor specification (AIR2216 II KV920).
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[15] Lumenier 6S 10,000 mAh 25C LiPo (XT90).
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Prices are list prices seen 5 November 2026. Performance figures are preliminary estimates derived from these sources.